

Fakulti: FAKULTI KEJURUTERAAN ELEKTRIK	
Nama Matapelajaran: Makmal Umum Tahun 3	Semakan : 1
Kod Matapelajaran : SKEE 3732	Tarikh Keluaran : October 2025
	Pindaan Terakhir :
	No. Prosedur : PK-UTM-FKE-(0)-10



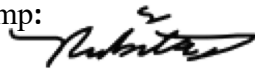
UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**FAKULTI KEJURUTERAAN ELEKTRIK
UNIVERSITI TEKNOLOGI MALAYSIA
KAMPUS JOHOR BAHRU
JOHOR**

SKEE 3732

MICROPROCESSOR LABORATORY

Laboratory 1: Introduction to STM32CubeIDE and GPIO

Prepared by: Dr Mohd Afzan Othman Signature & Stamp:  DR. MOHD AFZAN BIN OTHMAN Pensyarah Kanan Jabatan Elektronik dan Kejuruteraan Komputer Fakulti Kejuruteraan Elektrik Universiti Teknologi Malaysia 81310 Johor Bahru Johor, Malaysia Date: 2 October 2025	Approved by : Director of Programme Name: Prof. Ir Dr Rubita Sudirman Signature & Stamp:  PROFESOR IR. DR. RUBITA BINTI SUDIRMAN Head of Department Department of Electronic and Computer Engineering Faculty of Electrical Engineering Universiti Teknologi Malaysia 81310 UTM Johor Bahru, Johor Date: 2 October 2025
--	---

I. PRELIMINARY PREPARATION (15 marks)

Important Note: You are required to do following BEFORE the lab session.

1. Install STM32CubeIDE by downloading it from <https://www.st.com/en/development-tools/stm32cubeide>
2. Create a new project and compile the program (refer to Appendix A). Capture a screenshot showing "Build Finished" with no errors.
3. Using a suitable software (e.g., Proteus, Multisim, KiCad or EasyEDA), draw a complete circuit schematic based on the description in Table 1. Present the circuit to your supervisor for verification before proceeding with Lab 1.

Table 1: STM32 Connection

Device	STM32 pin	GPIO Port	Remarks
Switch0	PA0	GPIOA	Active HIGH
Switch1	PA1	GPIOA	Active HIGH
Switch2	PA2	GPIOA	Active HIGH
LED0	PB0	GPIOB	Common Cathode
LED1	PB1	GPIOB	Common Cathode
LED2	PB2	GPIOB	Common Cathode
LED3	PB3	GPIOB	Common Cathode
7-segment Display	PC0-PC7	GPIOC	Common Cathode

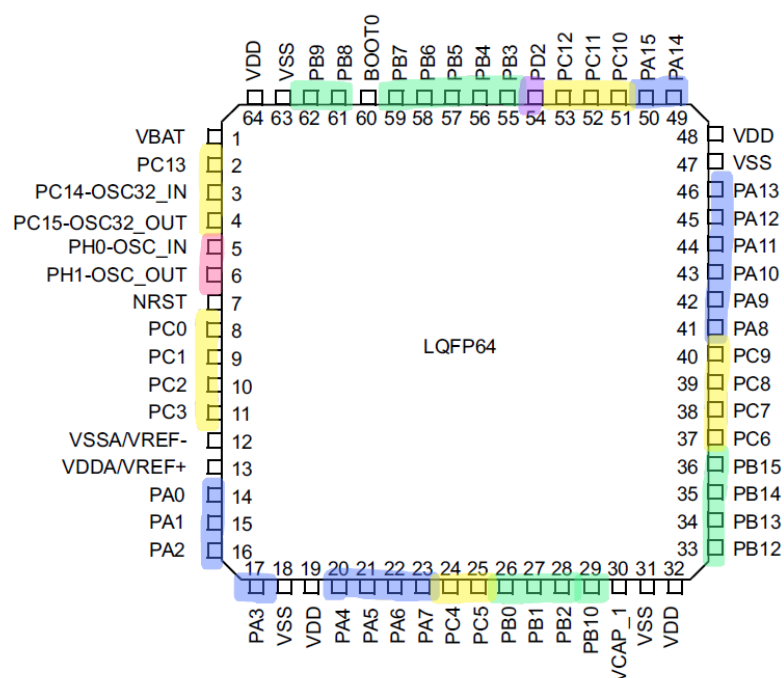


Figure 1: STM32F446RE Pin Out

II. LABORATORY SHEET

1 Title: GPIO Programming with Switches, LEDs, and 7-Segment Display

2 Objective:

At the end of this laboratory session, students will be able to:

1. Configure GPIO pins as input and output in STM32 bare-metal C programming.
2. Interface push button switches to control LED patterns.
3. Drive a 7-segment display to show the text.
4. Upload and run a program on STM32 Nucleo 64 board using STM32CubeIDE.

3 Equipment/Software/Reference:

1. STM32 Nucleo-64 F446RE development board
2. Breadboard and jumper wires
3. 3 push button switches with pull-down resistors (10 k Ω)
4. 4 LEDs with current-limiting resistors (220 Ω)
5. 1 seven-segment display (common cathode) with resistors (330 Ω)
6. USB cable for programming/debugging
7. Computer with **STM32CubeIDE** installed

4 Background Theory

The General Purpose Input/Output (GPIO) peripheral is the simplest method of communication between a microcontroller and external devices. A pin can be configured as input (to sense signals such as switches) or output (to drive devices such as LEDs or display segments).

In STM32, enabling GPIO requires:

- Activating the port clock in RCC_AHB1ENR.
- Setting the mode in GPIOx_MODER (input or output).
- Reading pin state from GPIOx_IDR (for switches).
- Writing values to GPIOx_ODR (for LEDs and displays).

In this lab, three switches act as input controls. Each switch triggers a unique LED pattern. When no switch is pressed, a seven-segment display shows the words. This experiment reinforces direct register-level programming (bare-metal) to understand how STM32 hardware works without HAL libraries.

5 Procedures

Note: You must complete Preliminary Preparation before proceeding this section. Make sure that STM32CubeIDE has been installed on your PC or Laptop.

1. Connect the STM32 Nucleo-64 board with all components listed in Table 1, based on the circuit designed during the pre-lab session. Refer Figure 2 for the board pins out connection. Present the completed circuit to the supervising lecturer for verification before uploading the program.
2. Open the project that was created during the pre-lab session.
3. Program the STM32 Nucleo-64 board and record the results and observations in Table 2.
4. Modify the program to execute the pattern sequence specified in Table 3 when SW1 and SW2 are pressed. Run the modified program, then present the working program to the supervising lecturer for endorsement.

Table 2: Result to be taken in procedure 3

Input	Describe Display at Seven Segment Panel	Describe Pattern generated at LED Panel
No Push Button pressed		
SW0 pressed & hold		
SW1 pressed & hold		
SW2 pressed & hold		

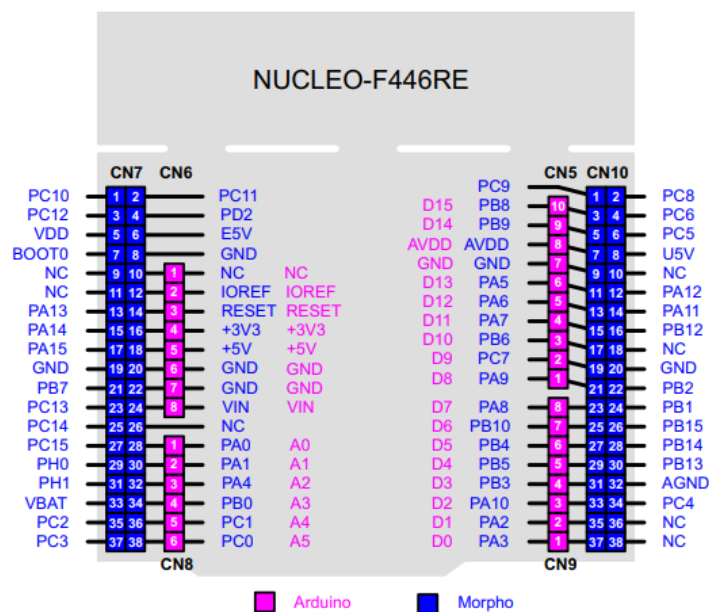


Figure 2: Nucleo 64 (F446RE) Board Pin Out

Table 3: Requirement for program modification

Switch Pressed	LEDs panel display requirement when respective button is selected	Remarks																																								
No switches pressed	Seven Segment LED display will display: S K E E 3 7 3 2 – <Group code> *for example: if you are section 5 group 1, thus: S K E E 3 7 3 2 – 5 0 1	Time delay for each blinking is 800ms																																								
SW1	Pattern1 <table><tr><th>State #</th><th>Led3</th><th>Led2</th><th>Led1</th><th>Led0</th></tr><tr><td>0</td><td>●</td><td>●</td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td>●</td><td>●</td></tr></table> Bold indicate LED ON	State #	Led3	Led2	Led1	Led0	0	●	●			1			●	●	Time delay for each blinking is 400ms																									
State #	Led3	Led2	Led1	Led0																																						
0	●	●																																								
1			●	●																																						
SW2	Pattern2 <table><tr><th>State #</th><th>Led3</th><th>Led2</th><th>Led1</th><th>Led0</th></tr><tr><td>0</td><td>●</td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td>●</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td>●</td><td></td></tr><tr><td>3</td><td></td><td></td><td></td><td>●</td></tr><tr><td>4</td><td></td><td></td><td>●</td><td></td></tr><tr><td>5</td><td></td><td>●</td><td></td><td></td></tr><tr><td>6</td><td>●</td><td></td><td></td><td></td></tr></table> Bold indicate LED ON	State #	Led3	Led2	Led1	Led0	0	●				1		●			2			●		3				●	4			●		5		●			6	●				Time delay for each blinking is 400ms
State #	Led3	Led2	Led1	Led0																																						
0	●																																									
1		●																																								
2			●																																							
3				●																																						
4			●																																							
5		●																																								
6	●																																									

Write the additional code and its location in the program that need to be made to implement Procedure 4.

You may Cut and Paste from modified program (do NOT screenshot):

Execution of Command when SW1 pressed: Working/Not working

Execution of Command when SW2 pressed: Working/Not working

Name and Signature of Lecturer:

6 Report Writing

Title:

Objective:

Equipment/Software Used:

Procedure:

Written in third person (reporting) speech:

Result:

Fill Table and Figures for Result and attach with result.

Discussion:

Discuss based on Objective and Result. There shouldn't be any alien Objective and Result.

Conclusion:

Conclude based on Objective, Result and Discussion.